

Area 17

1) a) $L = 0$ $ml = -1, 0, 1$
 $L = 1$

b) $L = 0, 1, 2$
 $m = -3, 0, 3$

2) $1s^2 2s^2 2p^6 3s^2 = \text{Magnesio}$

$$1s^2 = (1, 0, 0, 1/2) (1, 0, 0, -1/2)$$

$$2s^2 = (2, 0, 0, 1/2) (2, 0, 0, -1/2)$$

$$2p^6 = (2, 1, -1, 1/2) (2, 1, -1, -1/2)$$

$$(2, 1, 1, 1/2) (2, 1, 1, -1/2)$$

$$(3, 0, 0, 1/2) (3, 0, 0, -1/2)$$

$$3s^2 = (3, 0, 0, 1/2) (3, 0, 0, -1/2)$$

3) $Ca = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

$$Br = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$$

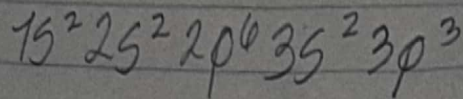
$$Se = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$$

$$Ca^{+2} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$$

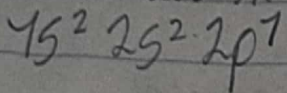
$$Sr^{+2} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$$

$$Zn = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$$

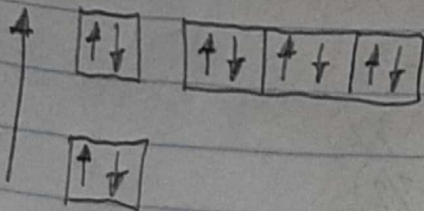
4) a) Al: al nivel 2p le hacen falta 2, es decir tiene que ser 6.



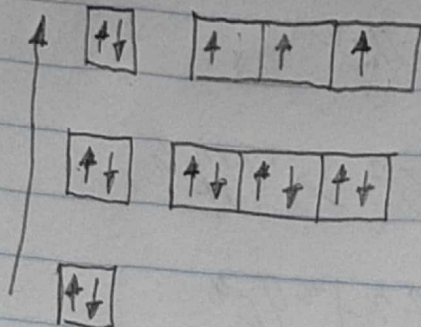
b) B: al sumar se pasa, en total debe ser 5



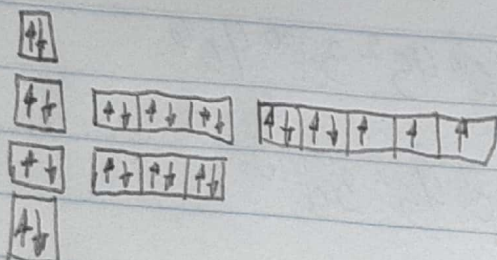
5) a) $1s^2 2s^2 2p^6$



b) $1s^2 2s^2 2p^6 3s^2 3p^3$



c) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$



$$6) S^{+6} = [\text{Ne}] 1s^2 2s^2 2p^6$$

$$1s^2 2s^2 2p^6 3s^2 3p^4$$

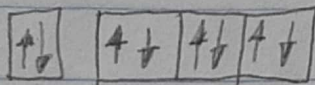
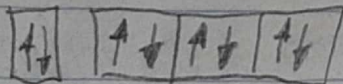
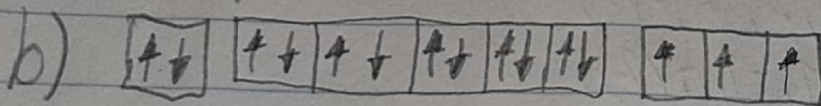
$$S^{+4} = [\text{Ne}] 3s^2$$

$$1s^2 2s^2 2p^6 3s^2$$

$$S^{-2} = [\text{Ne}] 3s^2 3p^6$$

$$1s^2 2s^2 2p^6 3s^2 3p^6$$

7) a) Es incorrecto porque los spin no se distribuyen 1 por 1 en una sola dirección.



$$\begin{aligned}
1s^2 &= (1, 0, 0, \frac{1}{2}) (1, 0, 0, -\frac{1}{2}) \\
2s^2 &= (2, 0, 0, \frac{1}{2}) (2, 0, 0, -\frac{1}{2}) \\
2p^4 &= (2, 1, -1, \frac{1}{2}) (2, 1, -1, -\frac{1}{2}) \\
&\quad (2, 1, 0, \frac{1}{2}) (2, 1, 0, -\frac{1}{2}) \\
&\quad (2, 1, 1, \frac{1}{2}) (2, 1, 1, -\frac{1}{2}) \\
3s^2 &= (3, 0, 0, \frac{1}{2}) (3, 0, 0, -\frac{1}{2}) \\
3p^4 &= (3, 1, -1, \frac{1}{2}) (3, 1, -1, -\frac{1}{2}) \\
&\quad (3, 1, 0, \frac{1}{2}) (3, 1, 0, -\frac{1}{2}) \\
&\quad (3, 1, 1, \frac{1}{2}) (3, 1, 1, -\frac{1}{2}) \\
4s^2 &= (4, 0, 0, \frac{1}{2}) (4, 0, 0, -\frac{1}{2}) \\
3d^{10} &= (3, 2, -2, \frac{1}{2}) (3, 2, -2, -\frac{1}{2}) \\
&\quad (3, 2, -1, \frac{1}{2}) (3, 2, -1, -\frac{1}{2}) \\
&\quad (3, 2, 0, \frac{1}{2}) (3, 2, 0, -\frac{1}{2}) \\
&\quad (3, 2, 1, \frac{1}{2}) (3, 2, 1, -\frac{1}{2}) \\
&\quad (3, 2, 2, \frac{1}{2}) (3, 2, 2, -\frac{1}{2}) \\
4p^3 &= (4, 1, -1, \frac{1}{2}) (4, 1, -1, -\frac{1}{2})
\end{aligned}$$

El electron no se encuentra que hace falta para llegar a el

- 8) a) Se viola la regla de Hund, primero se escriben los de multiplicidad mayor.
- b) No se cumple la exclusión de Pauli.
- c) No cumplen la regla de Pauli.
- f) Se viola regla de Hund